

B. Tech Degree V Semester (Supplementary) Examination **June 2008**

ME 505 HEAT AND MASS TRANSFER *(2002 Scheme)*

Time : 3 Hours

Maximum Marks : 100

(Approved Data book may be permitted)

- I. (a) Explain the concept of overall heat transfer coefficient and derive an expression for the same for a plane wall. (8)
- (b) A slab with thickness L and constant thermal conductivity k generates energy at a constant rate q_0 W/m³. The boundary surface at $x = 0$ is insulated (adiabatic) and that at $x = L$ dissipates heat by convection with a heat transfer coefficient h into a fluid at a temperature T_∞ . Develop expressions for the temperature $T(x)$. Also calculate the temperature at the surfaces $x = 0$ and $x = L$ under the following conditions : $L = 1$ cm, $k = 20$ W/mK, $q_0 = 8 \times 10^7$ W/m³, $h = 4000$ W/m²K and $T_\infty = 373$ K. (12)
- OR**
- II. (a) Prove that the temperature distribution in a body at time t during a Newtonian cooling or heating is given by $(T(t) - T_\infty)/(T_0 - T_\infty) = e^{-BiFo}$. State the important assumptions. (8)
- (b) A large slab of aluminium at a uniform temperature of 473 K suddenly has its surface temperature lowered to 343 K. What is the total heat removed from the slab per unit surface area when the temperature at a depth 4 cm has dropped to 393 K? Assume for aluminium $\alpha = 8.4 \times 10^{-5}$ m²/s and $k = 215$ W/mK. (12)
- III. (a) Explain the significance of Reynolds number, Grashof number and Prandtl number in convection heat transfer. (8)
- (b) Water flows with a mean velocity of $u = 2$ m/s inside a circular pipe of inside diameter $D = 5$ cm. The pipe wall is maintained at a uniform temperature of 373 K by condensing steam on its outer surface. At a location where the fluid is hydro dynamically and thermally developed the bulk mean temperature of water is 333 K. Calculate the heat transfer coefficient using the Sieder and Tate equation and Dittus Boelter equations. (12)
- OR**
- IV. (a) Explain pool boiling and the different regimes of boiling with a figure. (8)
- (b) A fine wire having a diameter of 0.02 mm is maintained at a constant temperature of 327 K by an electric current. The wire is exposed to air at 1 atm and 273 K. Calculate the electric power necessary to maintain the wire temperature if the length is 50 cm. Use the expression $Nu = 0.675 Ra^{0.058}$. (12)
- V. (a) Explain the following (i) Kirchhoffs law (ii) Reciprocity relation (iii) Gray body. (9)
- (b) The configuration of a furnace can be approximated as an equilateral triangular duct which is sufficiently long that the end effects are negligible. The hot wall is maintained

(Turn Over)

at $T_1 = 900 \text{ K}$ and has an emissivity $\varepsilon_1 = 0.8$. The cold wall is at $T_2 = 400 \text{ K}$ and has an emissivity $\varepsilon_2 = 0.8$. The third wall is a reradiating zone for which $Q_3 = 0$. Calculate the net radiation heat flux leaving the hot wall. Take $F_{1-2} = F_{1-3} = F_{2-3} = 0.5$. (11)

OR

- VI. (a) Explain the network analysis of radiation heat transfer problem. (8)
- (b) Two large parallel plates at $T_1 = 800 \text{ K}$ and $T_2 = 600 \text{ K}$ have emissivities $\varepsilon_1 = 0.5$ and $\varepsilon_2 = 0.8$, respectively. A radiation shield having an emissivity $\varepsilon_{3,1} = 0.1$ on one side and an emissivity $\varepsilon_{3,2} = 0.05$ on the other side is placed between the plates. Calculate the heat transfer rate by radiation per square meter with and without the radiation shield. (12)

- VII. (a) What is meant by fouling? How it affects the performance of heat exchangers? Derive an expression for the overall heat transfer coefficient with fouling on both sides of the tube. (8)
- (b) A one shell pass two tube pass heat exchanger has water on the tube side and engine oil on the shell side. It must be designed to heat 1.5 Kg/s water from 303 K to 353 K , with hot oil entering at 393 K and leaving at 353 K . The overall heat transfer coefficient is $250 \text{ W/m}^2 \text{ K}$. Calculate the heat transfer area required. Take specific heat of water = 4180 J/Kg K . (12)

OR

- VIII. (a) Explain $\varepsilon - NTU$ method for rating the heat exchangers. What is the significance of NTU on heat exchanger selection? (8)
- (b) A cross flow heat exchanger having an area 8.4 m^2 is to heat air [$c_p = 1005 \text{ J/Kg K}$] with water [$c_p = 4180 \text{ J/Kg K}$]. Air enters at 288 K and with a flow rate 2 Kg/s , while water enters at 363 K and 0.25 Kg/s . The overall heat transfer coefficient is $250 \text{ W/m}^2 \text{ K}$. Calculate the exit temperatures of both air and water as well as the total heat transfer rate. Consider both fluids unmixed on both sides of the heat exchanger. (12)

- IX. (a) Derive the Stefans equation for evaporation of water from a vessel. (10)
- (b) Two large vessels each containing uniform mixtures of nitrogen and carbon dioxide at 1 atm , and 288.9 K , but at different concentrations. Vessel 1 contains 90 mole percent nitrogen and $10 \text{ mole percent CO}_2$, whereas Vessel 2 contains 20 mole percent nitrogen and $80 \text{ mole percent CO}_2$. The two vessels are connected by a duct of diameter 0.15 m inside diameter and 1.22 m long. Determine the rate of transfer of nitrogen between the two vessels by assuming equimolar counter diffusion. The mass diffusivity for the nitrogen CO_2 mixture at 1 atm and 288.9 K can be taken as $0.16 \times 10^{-4} \text{ m}^2/\text{s}$. (10)

OR

- X. (a) Define (i) Mass flux (ii) Diffusion coefficient (iii) Mass transfer coefficient (iv) Sherwood number. (10)
- (b) A deep narrow cylindrical vessel is open at the top contains some toluene at the bottom. The entire system is at atmospheric pressure and 291.7 K . Under these conditions the diffusivity of air – toluene vapor is $0.826 \times 10^{-5} \text{ m}^2/\text{s}$, and the saturated vapor pressure of toluene at the liquid surface in the vessel is 0.026 atm . Determine the rate of evaporation of toluene into air per unit area of the liquid surface if the distance between the liquid toluene surface and the top of the vessel is 1.52 m . (10)

